FAQs - SFM-02

1. Stock prices follow which distribution?

Stock prices approximately follow the log-normal distribution, while stock returns approximately follow the normal distribution. A standard assumption in financial theory (and models like CAPM, Black-Scholes option pricing, portfolio theory) is that they strictly follow the above-specified distributions.

2. If we can know the movement of stocks w.r.t. each other through covariance, then what is the use of correlation?

The value of covariance is unbounded. Correlation is a standardized variation of covariance where the value lies between +/-1. So, while comparing bilateral relationships between different stocks, correlation is more helpful.

3. Isn't it better to take stock indices as the basis of the movement of other stocks for trading as opposed to individual stocks?

We can compare a stock with either its peer or the benchmark index. Peer comparison tells us how the stock is performing within the sector, whereas the latter tells us its performance with the index. Also, that is how beta is calculated generally for stocks.

4. Since the return is always positive, how does the standard deviation give an intuition of risk?

Returns can be negative. The standard deviation of returns tries to capture the extent of their dispersion. Based on the previously mentioned description, we conventionally take the standard deviation of a stock price or its returns to be positive.

5. What is R-square?

R-square measures how well the data fits the model specified. Its value lies between 0 and 1, the higher the better.

6. Why do we use N-1 in most of the statistical formulas?

We usually deal with sample data (and not the entire population data) while analyzing the phenomenon under observation. We extrapolate the characteristics of the population based on the sample data. However, certain biases creep in during the process. So, to correct for them, we use N-1 a.k.a Bessel's correction. Please read this or the Stats primer Section 2, Unit 7, Page 7 for more information.

7. What parameters do we need to consider while interpreting the linear regression output?

We usually look at the p-value of the variables that tells us whether the variable under consideration is statistically significant or not.

8. Can we calculate the variance of a portfolio consisting of more than two stocks?

Yes. It will be covered in the upcoming lecture. Meanwhile, you can refer to https://blog.quantinsti.com/calculating-covariance-matrix-portfolio-variance/

9. Is p-value a conditional probability?

Yes.

10. Correlation shows the direction of relationship and movement by positive and negative signs also but R-Square does not. So, isn't correlation better?

R-Square is used to just look at the goodness of fit, irrespective of the direction.